When Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: A Bilingual Diagnosis of Class-Conditional Negative Transfer

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Abstract

Sarcasm is non-literal language whose intended meaning may diverge from apparent sentiment, complicating detection in online text. As LLM-generated content appears alongside human writing, systems may need to judge both sarcasm and machine authorship. These judgments also interact in generation, where detector scores may be treated as measures of sarcastic intent and human authorship. Although sarcasm detection and machine-generated text detection (MGTD) are studied separately, their interaction in shared representations remains insufficiently understood. MGTD may promote source-predictive stylistic or distributional cues, whereas sarcasm detection should reflect pragmatic intent rather than source. We conduct, to our knowledge, the first bilingual diagnosis of how MGTD supervision affects sarcasm detection across Chinese and English. We harmonize bilingual resources, generate topic-matched AI texts, and build DualTest with source and sarcasm labels. Architecturally matched joint and single-task models reveal class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic. This pattern persists with a second generator and is attenuated by RCNN and interaction-projection modules. An exploratory comparison shows that detector scores for human authorship and LLM ratings of perceived human-likeness rank generated outputs differently. These findings motivate subgroup evaluation and caution against detector-only assessment.

1 Introduction

A sarcastic sentence typically conveys an attitude or intent that differs from, and may oppose, its literal meaning. Sarcasm can invert sentiment polarity and disrupt opinion-mining systems (Tay et al., 2018); ironic or sarcastic abuse is also a known challenge for content moderation (Vidgen et al., 2019). The task remains difficult because it depends on contrast, intent, and context. Annotation outcomes vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain and style (Jang and Frassinelli, 2024); multilingual settings introduce further variation (Casola et al., 2024) (§2.1).

LLM-generated text adds a source dimension to this task. Human-authored and AI-generated texts can each be sarcastic or non-sarcastic, so all four combinations are possible. A robust sarcasm detector should not infer sarcastic intent from source alone.

Machine-generated text detection (MGTD) imposes a contrasting representational demand: its objective distinguishes human-written from AI-generated text and therefore favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2). Some latent stylistic or distributional cues predictive of AI-generated text may overlap with, or become entangled with, features used for sarcasm detection. The output labels do not conflict; the potential conflict lies in how the two tasks organize and use the shared representation. Sarcasm detection should not rely on source differences alone, whereas MGTD must encode them. Shared-encoder multi-task learning can suffer task conflict and negative transfer (Mueller et al., 2022); more specifically, it can exploit knowledge useful for one task but spurious for another (Hu et al., 2022b). Evaluating the tasks only in separate models cannot reveal whether MGTD supervision changes the sarcasm decision.

We use joint training as a controlled intervention on the shared representation rather than as an assumed route to higher accuracy. A dual-head model lets us test how sarcasm decisions change when MGTD gradients also shape the shared representation. Architecturally matched single-head controls help isolate that change, and source × sarcasm subgroups show where it appears.

As a downstream implication, we examine whether the same detectors provide reliable evaluation signals for generated sarcasm. ViSP, for example, derives PPO rewards from a sarcasm detector (Wang et al., 2026), while generated sarcasm may be judged for both sarcastic intent and human-likeness (§2.4). If detector scores are source-sensitive, they could favor source-predictive stylistic or distributional cues rather than measure sarcastic intent. We therefore compare detector scores for human authorship with source-blinded LLM ratings of perceived human-likeness separately in §6.

We investigate the interaction through matched comparisons. We train dual-head models on bilingual Chinese–English data and build DualTest, whose dual labels partition evaluation into four source $\times$ sarcasm quadrants. Compared with matched single-head controls, joint training substantially increases the sarcasm false-positive rate on AI-generated text prompted to be non-sarcastic. We refer to this auxiliary-task-induced degradation, concentrated in a particular source $\times$ sarcasm subgroup, as class-conditional negative transfer. The pattern persists with a second generator and is attenuated in both RCNN and interaction-projection intermediate variants.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. Representation interventions show that the subgroup gap is sensitive to a dominant-variance subspace, while leaving the causal content of that subspace unresolved (§5.3.2, Appendix E). Because probe accuracy alone does not establish how encoded information is used (Hewitt and Liang, 2019; Elazar et al., 2021), we treat the source-correlated shortcut as a mechanism hypothesis with preliminary support rather than an established cause.

Contributions. To our knowledge, we provide the first bilingual diagnostic study of how MGTD supervision changes sarcasm detection across Chinese and English, together with DualTest for source $\times$ sarcasm analysis. Matched comparisons across models and generators identify class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic; cross-architecture comparisons characterize how its severity varies across configurations. Representation analyses characterize and constrain the source-shortcut hypothesis. Finally, an exploratory detector–judge comparison illustrates a downstream risk of using detector scores to evaluate generated sarcasm and perceived human-likeness.

2 Related Work

2.1 Sarcasm Detection and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. The task remains difficult in multilingual and class-imbalanced settings, especially when labels target intended rather than perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024). Although scores are not directly comparable across datasets, languages, modalities, or evaluation setups, they illustrate substantial benchmark variability. On English iSarcasmEval (Abu Farha et al., 2022), the top two systems report positive-class F1 of 0.605 and 0.569; across MultiPICo languages (Casola et al., 2024), zero-shot ChatGPT positive-class F1 ranges from 0.152 to 0.567. On the bilingual multimodal SarcNet benchmark (Yue et al., 2024), a text-only XLM-R baseline reaches 0.734 F1.

Architecturally, this work follows Yang and Ikeda (2024), who compared mBERT, mBERT-BiLSTM, and mBERT-RCNN on mixed English–Arabic and English–Chinese training sets and found that the RCNN variant generally improved performance on two English test sets. Their RCNN implementation builds on the recurrent convolutional architecture of Lai et al. (2015); we therefore include RCNN among the intermediate architectures examined in §4.1.

The benchmark results summarized above come from different datasets, so §5.1 provides directly comparable mBERT dual-head baselines on DualTest. To our knowledge, existing sarcasm resources do not provide both sarcasm and human-versus-AI source labels. DualTest supplies both labels in a bilingual Chinese–English setting.

The Chinese and English source resources differ in domain, annotation procedure, and class balance, so bilinguality introduces additional distributional variation. We treat language as a robustness and stratification dimension rather than as the primary transfer objective: the study asks whether the subgroup failure is confined to one language, not whether knowledge transfers across languages. We therefore report language-stratified results in addition to pooled metrics.

2.2 Machine-Generated Text Detection and Authorship Verification

Machine-generated text detection (MGTD) includes zero-shot model-likelihood methods, including probability-curvature detectors (Mitchell et al., 2023), and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and shows that detectors struggle on unseen domains or LLMs, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better cast as the latter. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

In our literature search, we found no work that jointly trains sarcasm detection and MGTD with a shared encoder.[1] This gap motivates our study because source information has different roles in the two tasks: it is the target of MGTD but should not by itself determine a sarcasm detector's prediction. Our study tests whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between task data (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a relevant NLP precedent through matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations can increase reliance on spurious correlations (Hu et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Mitigation under a shared encoder takes two broad forms. Parameter-efficient methods introduce task-specific or task-conditioned lightweight components (AdapterFusion, Pfeiffer et al., 2021; MAD-X, Pfeiffer et al., 2020; HyperFormer++, Karimi Mahabadi et al., 2021; PALs, Stickland and Murray, 2019). Other methods explicitly gate or adapt information sharing within the shared computation (Wu et al., 2019; CA-MTL, Pilault et al., 2021).

Rather than isolating task-specific parameters, we examine whether alternative intermediate representations alter the subgroup failure induced by joint training. We compare a no-module baseline with two intermediate-module variants (§5.3). Unlike aggregate negative transfer, our analysis asks whether auxiliary-task supervision degrades performance only for particular source $\times$ sarcasm subgroups.

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict studied above may also matter when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Model-based scores are also optimized as rewards, but overoptimization can increase a proxy score while degrading a gold-standard measure (Gao et al., 2023). A detector can achieve useful clas-

[1] Searches conducted in May 2026 and updated in July 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including "sarcasm detection," "machine-generated text detection," "AI-generated text detection," "multi-task learning," and "joint detection."

sification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. Multimodal sarcasm generation may face this risk: ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares detector scores for human authorship with separately obtained, source-blinded LLM ratings of perceived human-likeness.

3 Dataset Construction

Testing for interference requires separate task supervision and evaluation labels for authorship and sarcasm. Here "source" denotes human-versus-AI authorship; "corpus" denotes an input dataset. We combine iSarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None provides both labels. Mapping their annotations to a common binary sarcasm label yields 44,862 pooled instances (approximately 87% English), from which we construct three training sets and DualTest (Appendix A).

Sarcasm classification training set, SarcTrain (10,000, all human). Each language × label bucket contains 2,500 instances. Three buckets are downsampled; the Chinese sarcastic bucket is oversampled from 2,394 unique instances. The set is pre-split into five folds.

MGTD training set, AuthTrain (19,995). From 9,999 language-balanced human seeds, deepseek-r1:8b (DeepSeek-AI, 2025, via Ollama) produces 9,996 retained outputs (99.97% pairing coverage); Appendix A details quality-control exclusions.

Four language × style templates follow the balanced seed labels. Authorship and requested-style labels are constructed as separate, approximately balanced factors; no independence is assumed for realized textual features or perceived sarcasm (Appendix A).

Generation training set, GenTrain (2,428; used in §6). This set is drawn from two corpora containing topic or contextual information (iSarcasm and ToSarcasm; see Appendix A for their uses). Only positive-sarcasm instances are retained; approximately 40% are sampled, and the minority language is oversampled to 1,214 instances in each of Chinese and English. By prompt type, the set contains 374 rewriting, 839 response, and 1,215 comment instances, and serves as the instruction-tuning data for SFT.

Diagnostic test set, DualTest (4,664). The human subset combines official iSarcasm (1,400 English) and ToSarcasm (972 Chinese) test sets. Generation quality control retains AI counterparts for 2,292 of 2,372 texts (96.6% coverage). Human texts retain their annotations; for AI texts, the sarcasm label records the prompted condition rather than an independently annotated judgment of perceived sarcasm.

The four source × sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances. Appendix A gives the language × source × sarcasm distribution.

4 Dual-Head Detection Model

4.1 Tasks and Architecture

Given a Chinese or English input text x, the model produces two separate, non-mutually-exclusive binary predictions: whether x is labeled sarcastic and whether its authorship source is labeled human or AI. Any combination of the two labels is valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test whether MGTD training changes the representation read by the sarcasm head. This encoder was pretrained on English corpora; §5.1 directly compares it with multilingual-pretrained mBERT / XLM-R on the same bilingual test set. Encoder outputs are aggregated by an intermediate module and then passed to two task-specific classification heads; Figure 1 shows the main model.

The main model uses BiLSTM + CNN (RCNN) as its intermediate module. A single-layer bidirectional LSTM with hidden size 256 per direction produces a 512-dimensional output at each token position. We concatenate it with the 1,024-dimensional encoder output, then apply three groups of one-dimensional convolutions with kernel widths 3/4/5 and 128 channels each. Global max pooling produces a 384-dimensional vector z. After dropout, z is passed to the two heads. This module adapts the mBERT-RCNN intermediate ar-

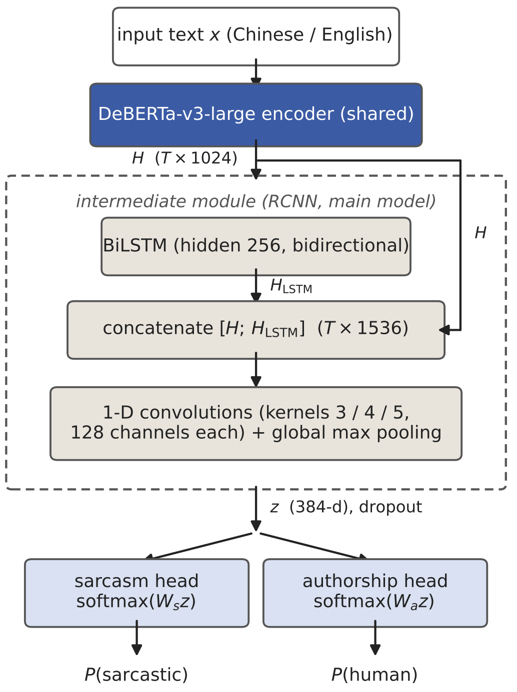

Figure 1: Main dual-head model: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with task-branch interaction projection, while retaining the same encoder backbone and task definitions (Appendix B).

chitecture of Yang and Ikeda (2024) to a DeBERTa-based dual-head setting.

To test the intermediate layer's role, §5 compares two variants. The plain model sends [CLS] directly to both heads. The interaction-projection model forms two learned 256-dimensional projections, h_s and h_a, and updates them as $h'_s = h_s + P_{a\to s}(h_a)$ and $h'_a = h_a + P_{s\to a}(h_s)$. Each branch contains one vector, so no query–key interaction remains; P is the effective learned value/output projection. The configurations retain the same encoder and task definitions while testing distinct intermediate-representation designs (Appendix B).

4.2 Training Protocol

Training uses a mixed loader that supplies task-homogeneous batches according to a 2:1 MGTD:sarcasm schedule, matching the approximate data-size ratio between AuthTrain and SarcTrain. Let the shared parameters be θ_{sh} (the encoder and intermediate module), let $z(x)$ be their aggregated representation for input x, and let W_s and W_a denote the two classification heads. At step t, the loss is computed only for the head corresponding to the task of the current batch B_t:

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|}\sum_{(x,y)\in B_t} \mathrm{CE}_w\big(W_s z(x), y\big), & B_t \subset \mathrm{SarcTrain}, \\ \frac{1}{|B_t|}\sum_{(x,y)\in B_t} \mathrm{CE}\big(W_a z(x), y\big), & B_t \subset \mathrm{AuthTrain}, \end{cases} \tag{1}$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B); because SarcTrain is class-balanced, the weighting emphasizes errors on sarcastic examples rather than correcting training-set imbalance. Softmax is applied to logits only when probabilities are required. The gradient flow follows from the equation: on sarcasm batches, $\partial\mathcal{L}/\partial W_a = 0$ (symmetrically, $\partial\mathcal{L}/\partial W_s = 0$ on MGTD batches), whereas θ_{sh} receives gradients from both types of batches. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than newly added modules. The sarcasm-head rate is fixed across configurations; MGTD-head rates are reported in Appendix B.

Model selection accounts for both tasks. At epoch e, the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}^{(e)}_{\text{stop}} = \frac{1}{2}\frac{\mathcal{L}^{(e)}_s}{\mathcal{L}^{(1)}_s} + \frac{1}{2}\frac{\mathcal{L}^{(e)}_a}{\mathcal{L}^{(1)}_a}, \tag{2}$$

where $\mathcal{L}^{(e)}_s$ and $\mathcal{L}^{(e)}_a$ are the epoch-e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics.

Five runs are defined by the SarcTrain validation folds: four folds train and one validates sarcasm, while AuthTrain uses a fixed 90/10 split. Each run reloads the pretrained encoder and randomly initializes new modules and heads. Configurations share fold assignments and nominal seeds, enabling split-matched comparisons but not identical initialization.

DualTest enters no training or validation split. Five-run standard deviations combine fold and initialization variation. Wilson intervals are conditional on a fixed final model and cover test-instance, not training or generator, uncertainty (Table 3; Appendix C.7).

Each subgroup configuration has a final model trained on all applicable data with a fixed epoch count determined from the five validation runs, an independent seed, and no early stopping. These models are not selected on DualTest or used in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

The experiments answer three questions in sequence. First, the backbone comparison selects a diagnostic model. Second, matched dual- and single-head models isolate the effect of adding MGTD training. Third, DualTest's labels divide the same predictions into four source $\times$ sarcasm quadrants to locate any effect hidden by overall F1. Aggregate comparisons report the mean $\pm$ standard deviation over five runs defined by SarcTrain validation folds. The quadrant analysis uses separately trained final models to obtain one prediction per instance for paired subgroup tests; Wilson intervals quantify binomial test-instance uncertainty conditional on each fixed model. No model is selected on DualTest.

5.1 Backbone Comparison

The backbone comparison selects a main model for the later diagnosis and provides mBERT baselines comparable to the prior work in §2.1. We compare dual-head models using mBERT, XLM-RoBERTa-base, DeBERTa-v3-base, and DeBERTa-v3-large. On the large encoder, we also compare the two intermediate-module variants from §4.1.

The encoders differ in pretraining. mBERT is BERT-base pretrained on multilingual Wikipedia (Devlin et al., 2019); XLM-RoBERTa-base uses CommonCrawl text from about one hundred languages (Conneau et al., 2020); and DeBERTa-v3 combines disentangled attention with ELECTRA-style pretraining (Clark et al., 2020) on primarily English corpora.

We report positive-class F1: the sarcastic class for sarcasm, following iSarcasmEval, and the human class for MGTD. Both metrics use the full DualTest set, aggregating languages and source sides. AI-side sarcasm labels follow the generation instructions (§3).

Table 1 reports the comparison (five-run mean $\pm$ standard deviation); the main model reaches sarcasm F1 0.6309 $\pm$ 0.020 and MGTD F1 0.9301 $\pm$ 0.005. Under the parameter scales and hyperparameter settings used here, DeBERTa-family encoders clearly outperform mBERT and XLM-R. The comparison mixes differences in parameter scale and pretraining corpora, and we use it only to pick the main model for the diagnosis.

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.004
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN (main model)	**0.6309** $\pm$.020	**0.9301** $\pm$.005

Table 1: Dual-head detection performance across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold = main model).

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126	0.6734	**−6.08**
Plain	MGTD	0.9221	0.9054	+1.67
RCNN	Sarc.	0.6309	0.6751	**−4.43**
RCNN	MGTD	0.9301	0.9253	+0.49

Table 2: Dual-head vs. single-head models (mean over five fold-based runs; Δ = dual − single, in percentage points).

RCNN has the highest mean, and the intermediate variants show 34–47% lower standard deviations across the five paired fold/seed runs.

Performance also differs by language. For the main model, Chinese sarcasm F1 is 0.819 and English sarcasm F1 is 0.429. The difference mainly reflects sarcasm prevalence (63.8% vs. 14.2%) and source-benchmark difficulty (Appendix C.7).

Across all models, sarcasm F1 is lower than MGTD F1. Appendix C reports macro-averaged, human-subset, and per-language results; the cross-dataset scores summarized in §2.1 are not directly comparable with DualTest.

5.2 Dual-Head versus Single-Head Ablation

This ablation asks whether the sarcasm classifier changes because of joint training rather than because of architecture or sarcasm data. Within each architecture, the dual- and single-head models share the encoder, intermediate structure, sarcasm data, and fold assignments. They differ only in whether the other task is included during training, and comparisons are paired run by run (§4.2).

Joint training lowers sarcasm F1 and slightly improves MGTD F1. Across the two architectures, all ten paired sarcasm differences are negative (-0.8 to -8.6 pp), whereas MGTD differences are centered around zero and are predominantly positive. This asymmetry does not depend on architecture.

RCNN moves both effects toward zero, consistent with partial buffering of inter-task interference; the quadrant analysis next examines the subgroup pattern.

5.3 Four-Quadrant Diagnosis: Class-Conditional Negative Transfer and Intermediate-Layer Buffering

Aggregate metrics cannot show whether an error depends on both source and sarcasm class. The overall sarcasm F1 of the plain dual-head model looks like unremarkable mid-range performance, although its errors concentrate on AI text. We therefore partition DualTest by its source $\times$ sarcasm labels and examine each combination separately. Human labels are dataset annotations, whereas AI labels record the prompted condition. Because each quadrant contains one label class, F1 is not informative within a quadrant: positive subsets contain no negative examples, and vice versa. We report recall for sarcastic subsets and the false-positive rate for non-sarcastic subsets.

5.3.1 Three Architectures and Single-Head Controls

We compare five detectors on the same DualTest set. The three dual-head variants (plain, interaction projection, and RCNN) show how the failure changes with intermediate structure. Each of the two single-head models matches its dual-head counterpart in architecture and sarcasm supervision; only the AuthTrain data and MGTD objective are removed.

The single-head models provide the counterfactual. If the failure came from training the sarcasm task only on human text, the matched single-head model should fail in the same way. If joint training induces the failure, the single-head model should not show it.

Table 3 reports the sarcasm predictions. Corresponding MGTD results are reported below and in Appendix C.

The plain dual-head model predicts sarcasm for 86.3% of AI $\times$ non-sarcastic instances. The error is therefore concentrated where AI-source cues co-occur with a non-sarcastic prompted condition, which is consistent with reliance on source-associated cues.

Both intermediate modules reduce the false-positive rate by more than 45 pp from the plain model, with non-overlapping confidence intervals (Table 3). RCNN is about 7 pp lower than interaction projection, and that smaller difference is also significant (Appendix C.7).

The matched single-head models have a false-positive rate of only 19.3%, despite also receiving sarcasm supervision only from human text. On the same instances, the difference from the plain dual-head model is significant by a paired McNemar test (discordant pairs 1,021 / 7, $p < 0.001$; exact values and other comparisons are in Appendix C.7). Missing AI text from SarcTrain therefore cannot explain the failure; joint training induces it.

The plain and RCNN single-head models also behave almost identically, both at approximately 19.3%. RCNN's mitigating effect thus appears only under joint training, not as general architectural robustness to AI text.

The MGTD head provides a control for the failure's directionality. On the same AI $\times$ non-sarcastic subset, its false-positive rate is only 6.1% (plain) / 6.6% (RCNN), and it remains stable across all four quadrants. The single-head MGTD counterfactual performs at the same level as the dual-head models (complete values in Appendix C). On the same set of AI texts, the problem is concentrated in the sarcasm head and the gap is on the order of tens of percentage points.

The failure depends jointly on source and sarcasm class and is most extreme for AI $\times$ non-sarcastic text. We therefore call it class-conditional negative transfer. The subgroup failure and the asymmetric aggregate transfer in §5.2 describe the same effect at different levels.

5.3.2 Mechanism Explanation and Levels of Evidence

Because MGTD gradients update the shared representation whereas SarcTrain contains only human texts, we hypothesize that the sarcasm head exploits source-correlated style cues that are not corrected by sarcasm supervision on AI text. Under this interpretation, an intermediate module may partially buffer changes in the shared representation. This account is a hypothesis rather than an established causal mechanism.

The representation evidence is mixed. Source is

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	**86.3%** [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	**40.9%** [38.5, 43.4]
RCNN dual-head (main)	80.8%	40.5%	89.3%	**34.1%** [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	**19.3%** [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	**19.3%** [17.4, 21.4]

Table 3: Five sarcasm classifiers on four source × sarcasm quadrants (recall / false-positive rate, FP). Human labels are annotations; AI labels record prompted conditions. Brackets give model-conditional Wilson 95% intervals; full intervals and paired tests are in Appendix C.7.

highly decodable and correlates with sarcasm decisions. LEACE (Belrose et al., 2023) reduces both source decodability and the AI × non-sarcastic false-positive rate.

However, removing a high-variance direction orthogonal to source produces a similar reduction while leaving source decodability intact; ten random-direction placebos leave the rate unchanged. The effective interventions therefore identify sensitivity to a dominant-variance subspace, but they do not establish source identity or any specific stylistic feature as causal. We accordingly use the term source-correlated shortcut. Appendix E reports the full probing, erasure, and control results.

5.3.3 Cross-Generator Replication

To test whether the failure depends on one generator's style, we regenerate the AI side of DualTest with qwen3.5:9b (Qwen Team, 2026). The human texts and trained detectors remain unchanged. Appendix C.7 plots both generators with model-conditional Wilson 95% intervals and gives the paired tests.

With qwen3.5:9b, the plain dual-head false-positive rate rises from 86.3% to 91.3%, while the matched single-head rate falls from 19.3% to 7.9%. Mitigation also persists for RCNN (34.1% → 22.1%) and interaction projection (40.9% → 29.5%). The second generator therefore reproduces the class-conditional failure, although broader generator families remain untested.

6 Exploratory Detector–Judge Misalignment

The preceding experiments treat the detector as a classifier. As a downstream implication, we now ask a related but separate question: whether its continuous scores track separately obtained, source-blinded judgments of generated text. This distinction matters because a classification boundary need not measure generation quality.

We score 600 base and SFT outputs from 300 prompts. We use the main detector and three blinded LLM judges: DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano (DeepSeek-AI, 2024; Gemini Team, 2025; OpenAI, 2026). The SFT model is Qwen3-8B (Qwen Team, 2025) fine-tuned on GenTrain with QLoRA (Dettmers et al., 2023); Appendices B–C detail the protocol.

The two score sources rank the outputs differently. In Chinese, SFT raises the detector's sarcasm score from 0.780 to 0.895 but lowers the mean judge score from 4.01 to 3.71. The MGTD head also assigns higher human-authorship probabilities to SFT outputs in both languages, whereas the judges' perceived human-likeness ratings move in the opposite direction. Pooled Spearman correlation is -0.212 between human-authorship probabilities and human-likeness ratings and -0.020 for Chinese sarcasm.

The judges are LLMs, not human annotators, so what these results establish is detector–judge misalignment; how human readers would rate the same outputs is a question we do not answer. Even so, detector scores and a separate evaluation signal rank the same outputs differently, and that alone argues against using detector scores as standalone generation rewards. The full protocol, per-judge tests, within-condition correlations, language analysis, and representative cases are reported in Appendices C–D.

7 Conclusion

Source and sarcasm labels can coexist without conflict, but training them through one representation can still change how the sarcasm decision is made. The plain dual-head model, matched to its single-head counterpart in architecture and sarcasm supervision, sharply overpredicts sarcasm on AI-generated text prompted to be non-sarcastic; the failure persists with a second generator, and two intermediate configurations reduce it. Representation analyses support a source-correlated shortcut hypothesis and show sensitivity to a dominant-variance subspace, while leaving causal attribution open. On the generation side, SFT produced a language asymmetry, and detector and judge scores ranked the same outputs differently, so neither MGTD nor sarcasm detector scores should be treated as standalone generation-quality signals. When a source-predictive objective shares a representation with a semantic judgment, overall metrics should be supplemented with source $\times$ class subgroup evaluation.

Limitations

DualTest combines original sarcasm annotations for human texts with instruction-conditioned style labels for AI texts. Because humans did not independently annotate the AI-side labels, absolute AI-side scores measure agreement with the requested style.

Replication with a second generator shows that the failure is not limited to deepseek-r1:8b, but it does not establish generalization across broader generator families. The input corpora also operationalize sarcasm differently, and the model does not use conversational context.

The mechanism evidence is primarily behavioral and linear. Source information is decodable and correlated with sarcasm decisions, yet the orthogonal high-variance control blocks causal attribution to source identity. The architecture comparison provides convergent configuration-level evidence for buffering, but it is not capacity-matched and the MGTD-head rate differs for intermediate variants. Crucially, the dual–single comparisons within each architecture use the same sarcasm data, class weights, and optimization settings, keeping these factors matched when MGTD supervision is added; an unweighted-loss control could further characterize the effect's absolute severity.

The generation analysis uses three blinded LLM judges rather than human evaluators and has a 60:40 Chinese–English sample ratio. Chinese and English prompts also instantiate different generation tasks, so cross-language differences conflate language with prompt and task format. The analysis therefore establishes a directional disagreement between detector and judge scores, not human-perceived quality or a pure language effect.

We also do not test reward-based generation. Future work should measure how the observed misalignment affects reinforcement learning, apply causal-mediation and nonlinear tests to source attribution, and evaluate parameter-isolating or data-side mitigations.

Ethical considerations

All human-authored texts come from publicly released research datasets: iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. We use them under their original licenses, collect no new user data, and process no personally identifiable information.

All AI texts were generated locally by open-source models (deepseek-r1:8b and qwen3.5:9b). They are used only for training and evaluation and are labeled as AI-generated. The blinded evaluation uses automated API calls to DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano; it involves no human participants.

The detectors are developed for research diagnosis. We explicitly warn (§6) that their scores should not be used directly as generation-quality or reward signals, nor should the detectors be deployed in high-risk settings such as content moderation without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here (the systematic prediction of sarcasm for AI-generated text prompted to be non-sarcastic) is itself an example of the systematic bias such systems can direct at particular content subgroups.

AI-assisted tools were used during writing for language polishing and formatting; the study design, experiments, and conclusions are the authors' own, and the authors take full responsibility for the work.

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A Detailed Dataset Statistics and Quality Checks

A.1 Corpus details. The pooled corpus collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance in SarcTrain is achieved through bucketed

Corpus	Lang.	Inst.	Use
iSarcasm (SemEval-2022 T6)	English	3,467	Sarc. train + SFT
CSC	English	6,878	Sarc. train
ToSarcasm	zh 3,892 / en 6	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Input corpora. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

sampling. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic bucket contains 2,394 available unique instances and is oversampled within the bucket to reach 2,500, producing the 4.24% duplication rate for that bucket reported in A.3.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language × source × sarcasm distribution of DualTest. Human labels are dataset annotations; AI labels record prompted conditions.

A.2 Three-dimensional distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds and find 99.97% completeness of AuthTrain human–AI pairing (9,996 / 9,999 seeds). Exact-match screening finds two SarcTrain instances (0.02%) and five AuthTrain instances (0.025%) that also occur on the AI side of DualTest; overlap on the human subset is zero. The seven matches are negligible relative to the training and evaluation set sizes and are disclosed for completeness. SarcTrain's text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1). Post-processing also checks for chain-of-thought (Wei et al., 2022) leakage.

A.4 AI generation protocol. All AI texts are produced by locally deployed generation models through the Ollama API (deepseek-r1:8b for the main results; qwen3.5:9b for cross-generator replication), using temperature $= 0.8$, top-$p = 0.95$ (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four templates defined by language × style, with style determined by the seed instance's sarcasm label. The templates explicitly constrain output style and length (20–80 Chinese characters; 10–50 English words); the length constraint also mitigates the confound of distinguishing human and AI texts by length (see the length-controlled analysis in Appendix E). The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

```
【任务】生成一条讽刺风格的评论
【主题】{seed}
【要求】
- 直接给出简短的评论，不要解释思路或分析
- 风格要讽刺、幽默、夸张
- 表达与主题相关的观点即可，不必追求严谨
- 20-80 字为佳
【评论】
```

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fěngcì fēnggé de pínglùn
[zhǔtí] {seed}
[yāoqiú]
- zhíjiē gěichū jiǎnduǎn de pínglùn, bùyào jiěshì sīlù huò fēnxī
- fēnggé yào fěngcì, yōumò, kuāzhāng
- biǎodá yǔ zhǔtí xiāngguān de guāndiǎn jíkě, bùbì zhuīqiú yánjǐn
- 20–80 zì wéi jiā
[pínglùn]

Its English translation is:

```
[Task] Generate a sarcastic-style comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment; do not explain your reasoning or analysis
- The style should be sarcastic, humorous, and exaggerated
- It is sufficient to express a viewpoint related to the topic; strict rigor is not required
```

```
- Preferably 20--80 Chinese
characters
[Comment]
```

The Chinese non-sarcastic template changes only the style line to “The style should be normal and natural.” The English templates follow the same structure, using either “sarcastic, humorous, exaggerated” or “normal, natural,” with a length limit of 10–50 words.

Post-processing removes <think>…</think> segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ń* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that remain overlength or cyclic after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The cross-generator replication has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from deepseek-r1:8b.

A.5 SFT training and generation-evaluation set. GenTrain (2,428; 1,214 each in Chinese and English) is constructed by sampling approximately 40% of positive-sarcasm instances suitable for generation tasks and oversampling the minority language to achieve balance. By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. The detector scoring and three source-blinded LLM evaluations in §6 are both conducted on this set.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	MGTD
Plain / backbone	5e-7	—	5e-6	1e-5
Intermediate variants	5e-7	5e-6	5e-6	5e-6
Single-head	5e-7	5e-6†	5e-6	1e-5

Table 6: Discriminative learning rates. †RCNN single-head only.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5e-6 across configurations. The MGTD-head rate is 1e-5 for the plain dual-head and single-head models and 5e-6 for the intermediate variants. This difference is reported for completeness and is considered when interpreting cross-configuration effects; the matched dual–single comparisons retain the same settings within each architecture.

B.2 Shared training hyperparameters. Training uses at most 10 epochs, 200 warmup steps, weight decay 0.01, early-stopping patience 5, and gradient clipping 1.0. The physical batch size is 4 with gradient accumulation 4, giving an effective batch size of 16. The MGTD:sarcasm sampling ratio is 2:1, and sarcasm class weights are [1.0, 2.5]. Both choices were specified before testing. Because SarcTrain is balanced, the class weights implement cost-sensitive emphasis rather than imbalance correction. The normalized joint validation loss is $0.5\times$ [sarcasm validation loss / first-epoch value] $+0.5\times$ [MGTD validation loss / first-epoch value].

All fold-based runs start from seed 42 plus the fold index and share the same fold assignments and nominal seed scheme across architectures and ablations. MGTD data use a fixed 90/10 holdout in every fold. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs. The final model uses all applicable training data, seed $+1000$, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

Software and hardware environment. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB); PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); blinded LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). We fine-tune Qwen3-8B with 4-bit loading and bfloat16 computation. LoRA (Hu et al., 2022a) uses rank 64, alpha 16, dropout 0, and targets q/k/v/o_proj plus gate/up/down_proj. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses learning rate 5e-5, three epochs, batch size 1, gradient accumulation 8, and warmup ratio 0.05. Validation loss selects the checkpoint.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Main-model macro-averaged F1, under the same five-run protocol and calculated from each run's confusion matrix: sarcasm **0.6678** $\pm$ **0.036** and MGTD **0.9298** $\pm$ **0.005**. Macro-averaged F1 is higher than positive-class F1 (0.6309) because the majority class has a higher F1; the ordering in §5.1 remains unchanged under both conventions.

C.2 Complete four-quadrant results for the MGTD head (including single-head counterfactuals). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	**H×s**	**H×n**	**AI×s**	**AI×n**
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual (main)	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head MGTD counterfactuals remain within single-digit percentage points of the dual-head models in every quadrant. Joint training therefore induces no class-conditional failure on the MGTD side, consistent with §5.3.

The MGTD head is also stable on the qwen3.5:9b replication set. AI $\times$ non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN. AI $\times$ sarcastic rates are all at most 0.4%, and overall MGTD F1 scores are 0.952, 0.917, and 0.966.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Each sample is sent separately to the three judge APIs with one prompt (May 2026; temperature $=$ 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. Base and SFT outputs use the same prompt and are evaluated separately.

Pairwise instance-level agreement is moderate to low. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions, not agreement in instance rankings.

Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that SFT receives lower sarcasm ratings than the base model in five of six language $\times$ judge groups. The pooled correlations in Table 8 use the mean of the three judges, partly averaging inter-judge noise.

Subset	n	**Sarc.** ρ	**MGTD** ρ
Overall	600	$+0.136^*$	$\mathbf{-0.212}^*$
Chinese	360	$\mathbf{-0.020}$	-0.218^*
English	240	$+0.338^*$	-0.270^*

Table 8: Spearman correlations between detector scores and source-blinded LLM judgments (main-model detector, $n = 600$ pooled over base and SFT). * significant; other entries not significant.

Judge / lang.	n	**SFT**	**Base**	**adj.** p
DeepSeek-V3 / en	120	3.292	4.025	$<.0001$
DeepSeek-V3 / zh	180	3.878	4.044	.0008
Gemini-2.5-FL / en	120	3.750	3.992	.0008
Gemini-2.5-FL / zh	180	3.994	4.028	.2008*
GPT-5-nano / en	119	3.681	4.084	$<.0001$
GPT-5-nano / zh	180	3.517	3.911	$<.0001$

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). *not significant.

C.4 Paired tests between SFT and human references (supplementary). This test is performed on the Chinese subset of the generation-evaluation set ($n = 180$; GPT-5-nano is missing three instances, leaving 177). Directions are inconsistent: for DeepSeek-V3, SFT is significantly below the reference on human-likeness (-0.161, $p = 0.0025$); for GPT-5-nano, SFT is significantly above the reference on sarcasm ($+0.186$, $p = 0.0114$); the other four groups are not significant.

C.5 Plain-detector control (robustness). With the plain detector, Chinese sarcasm changes from 0.765 to 0.763 (-0.001), English sarcasm from 0.843 to 0.659 (-0.184), and the MGTD score rises by $+0.760$ in Chinese and $+0.705$ in English. The two detectors therefore agree on the rise in MGTD scores, while the sarcasm direction depends on the detector.

Correlations are $+0.142$ for sarcasm and -0.214 for MGTD overall; -0.037 (not significant) and -0.209 in Chinese; and $+0.342$ and -0.283 in English. Each differs from the main-model value by at most 0.02. The detector–judge misalignment is therefore robust to detector architecture.

Within-condition correlations (main-model detector). For MGTD, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$, English -0.09). The SFT condition has $\rho = -0.12$ ($p = 0.039$; Chinese -0.19, $p = 0.009$; English -0.12, not significant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises the detector score by about 0.67 while lowering the mean blinded rating. There is no positive detector–judge alignment on this dimension within conditions.

For sarcasm, within-condition correlations are $+0.05$ for Chinese base (not significant), $+0.15$ for Chinese SFT ($p = 0.048$), $+0.41$ for English base, and $+0.37$ for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.6 Detector scores before and after SFT and language asymmetry. Scores from the main-model detector for base and SFT outputs (0–1):

Lang.	Cond.	Sarc.	MGTD
Chinese	Base → SFT	0.780 → **0.895**	0.289 → 0.969
English	Base → SFT	0.854 → **0.730**	0.198 → 0.864

Table 10: Detector scores before/after SFT (main-model detector).

Sarcasm scores move in opposite directions by language: $+0.115$ in Chinese and -0.124 in English. The 60:40 language imbalance in SFT training may partly explain this difference (see Limitations). MGTD scores rise sharply in both languages, so the detector assigns SFT outputs higher probabilities of human authorship.

Mean LLM judgments for sarcasm / human-likeness are 3.77/4.17 for human references, 4.01/4.24 for the base model, and 3.71/4.08 for SFT. These directions conflict with the detector's MGTD score, as analyzed in §6.

C.7 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1. Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

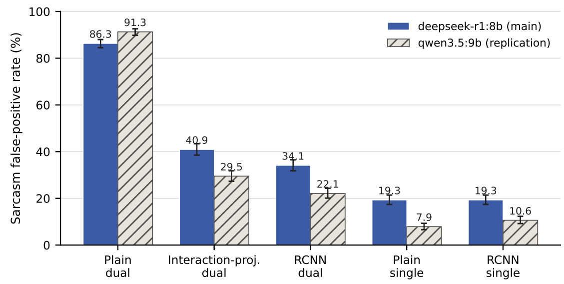

Figure 2: Sarcasm false-positive rates on the AI × non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

The five detectors are trained independently, yet their false-positive-rate ordering is stable and replicates with the second generator. Between-run variation in overall F1 (±0.02–0.03; Table 1) is also much smaller than the between-quadrant differences.

Model	deepseek	qwen
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual (main)	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI × non-sarcastic subset (two generators; %).

Intervals do not overlap between any dual-head and single-head model, or between the plain dual-head model and either intermediate dual-head model. Exact paired McNemar tests use the same AI × non-sarcastic texts from the main-generator set. Discordant pairs are 1,021 / 7 for plain dual vs. plain single ($p < 10^{-291}$), 239 / 15 for RCNN dual vs. RCNN single ($p < 10^{-52}$), 805 / 15 for plain dual vs. RCNN dual ($p < 10^{-215}$), and 203 / 100 for interaction projection vs. RCNN dual ($p = 3.3 \times 10^{-9}$). All four differences are significant.

Main-model MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the source label is constant within the human subset, positive-class F1 reduces to recall and is not reported separately.

Chinese sarcasm F1 is higher than English sarcasm F1 for the main model (0.819 vs. 0.429). Two data properties explain much of the difference. Sarcasm prevalence is 63.8% in Chinese but 14.2% in English, making positive-class F1 values difficult to compare directly. The source benchmarks also differ: the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval, where the best systems reach about 0.56–0.61 F1 (§2.1), whereas the Chinese side uses topic-comment-style ToSarcasm.

The encoder does not provide an obvious alternative explanation. DeBERTa-v3's SentencePiece vocabulary (Kudo and Richardson, 2018) tokenizes Chinese largely by character, at approximately 1.02 tokens per character with 2.5% out-of-vocabulary symbols. Chinese F1 remains high under this tokenization, consistent with prevalence and genre playing the larger roles.

The main model's overall sarcasm F1 is 0.663, compared with 0.629 on human texts alone, because the overall score also includes AI texts (§5.1). Within the human subset, dual–single differences are below 0.03 and vary in direction. The visible negative-transfer loss is therefore concentrated in the AI-side quadrants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore shifts across the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sarcasm scores while largely preserving their internal ordering. This pattern is consistent with a source-related component being added to the sarcasm signal.

D Representative Cases of Detector–LLM Score Misalignment

Cases are selected from the paired table of main-model detector scores and blinded judgments from the three LLMs by the magnitude of disagreement (detector scores: 0–1; LLM scores: mean of three evaluators on a 1–5 scale; long texts truncated). Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

D.1 Classified as human-authored by the detector but rated as less human-like by the LLMs (MGTD-score misalignment; corresponding to the analysis in §6). In the first case, AI-generated text with obvious chain-of-thought traces is nevertheless classified as human by the MGTD head with confidence 1.00 (see §6 for the mechanism interpretation).

D.2 Classified as "sarcastic" by the detector but "not sarcastic" by the LLMs.

D.3 Classified as "sarcastic" by the LLMs but missed by the detector. Together, D.2 and D.3 suggest that the sarcasm head favors surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences. It is less responsive to complete written irony that depends on semantic reversal, while the LLM judges show the opposite tendency.

The cases were selected by disagreement magnitude, so they do not estimate how often either pattern occurs. They provide qualitative examples of the near-zero Chinese sarcasm correlation in §6; systematic testing remains future work.

Model	Full set	Chinese	English	Human subset	Human·zh	Human·en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head (main)	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 12: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for the main model).

L	S	P_{hum}/LLM	Text (excerpt)
zh	SFT	1.00 / 2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [*hǎo, wǒ xūyào zhēnduì Sānxīng huíduì Huáwéi de xīnwén shēngchéng yī tiáo fěngcì pínglùn. shǒuxiān…*] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First…”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [*pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā*] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 13: MGTD-score misalignment cases. $P_{\text{hum}} = P(\text{human})$.

L	S	P_{sar}/LLM	Text (excerpt)
zh	SFT	1.00 / 2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [*Sānxīng: wǒ cái shì dì-yī! nǐ kànkan wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!*] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67	“恭喜恭喜，美国牛牛牛，继续加油” [*gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu*] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
en	SFT	0.93 / 2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 14: Sarcasm false-positive cases. $P_{\text{sar}} = P(\text{sarcastic})$.

L	S	P_{sar}/LLM	Text (excerpt)
zh	Base	0.14 / 4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [*Tèlǎngpǔ chéngnuò 120 yì bǔtiē xiàng sǎ dàbǐng, nóngmín shōudào de què shì suōshuǐ chéng yìngbì de chéngnuò…*] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins…”] (metaphorical irony)
zh	Base	0.24 / 4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [*Tèsīlā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiánbāo tiàoshuǐ hòu yòu tiàoshuǐ.*] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis + ironic emoji)

Table 15: Sarcasm false-negative cases. $P_{\text{sar}} = P(\text{sarcastic})$.

E Preliminary Representation-Level Analysis of the Failure Mechanism

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The interpretation of conclusions follows §5.3.2.

E.1 Setup. All analyses freeze the plain dual-head model that produces the 86.3% main-text result. Direction estimation and evaluation are cross-fitted: one half estimates the direction and the other computes metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

Implementation checks pass. Manually computed logits match model outputs with maximum error 1.4×10^{-6}, and the erasure-projection residual is about 10^{-6}. LEACE reduces fitting-half source AUC from 0.988 to 0.042, as its definition requires. Probe selectivity is 0.986 with true labels and 0.512 with permuted labels.

E.2 Protocol. The linear probe is logistic regression with five-fold cross-validation. A selectivity control retrains it with permuted source labels, and correlation analyses are repeated within text-length tertiles. Alignment is the directional cosine in whitened geometry ($\Sigma^{1/2}w$).

Erasure tests cover one direction, INLP subspaces ($k = 1$–16; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Placebos use ten random whitened directions and directions estimated from permuted labels. srcORTH-PC1 is the maximum-variance direction after removing its projection onto the source direction. Erasing it leaves source decodability unchanged, separating perturbation of dominant variance from removal of source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also gives 0.974, so source decodability is not unique to joint training. On AI $\times$ non-sarcastic texts, source and sarcasm logits correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry (cos ≈ -0.03) but moderately aligned after whitening. The absolute cosine is 0.41 using the probe direction and 0.49 using the MGTD-head direction. These results are compatible with the shortcut hypothesis, but remain correlational.

E.4 Intervention evidence. gap $=$ AI $\times$ non-sarcastic false-positive rate $-$ human $\times$ non-sarcastic false-positive rate; source AUC is the decoding ability of a probe retrained after erasure.

Erasure scheme	AI×n FP	H×n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (full)	68.7	49.5	19.1	**0.589**
Random whitened ×10	84.9	32.7	—	0.982
srcORTH-PC1	**66.7**	54.1	12.6	**0.982**

Table 16: Intervention evidence. H×n = human × non-sarcastic; AI×n = AI × non-sarcastic; FP in %.

Single-direction and low-dimensional INLP erasure are ineffective. LEACE initially appears effective: it erases source information, reduces the gap from 52.0 to 19.1, and outperforms all random placebos.

The orthogonal high-variance control changes the interpretation. It leaves source decodability intact but reduces the false-positive rate to 66.7 and the gap to 12.6, an effect similar to LEACE. The intervention therefore identifies a high-variance subspace without isolating source identity as the cause.

E.5 Supplementary observation. The mean false-positive rate across the two non-sarcastic source sides is almost unchanged before and after the intervention (58.8 $\rightarrow$ 59.1). The intervention does not "fix" the model; it merely transfers part of the overprediction of sarcasm on the AI side to the human side.